

Semantic Similarity in Latent Space for LLM Prediction Evaluation

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Main Finding

Embedding similarity can serve as a threshold-based correctness indicator only after refinement. Raw cosine is near random (AUROC **0.48–0.53**); unified refined cosine reaches **0.76–0.93** AUROC, and task-specific scoring gives the best F1: short hybrid for short QA and claim/chunk alignment for long QA.

1. Introduction & Background

Problem: LLM outputs can be fluent but factually wrong.

Why hard: exact match is strict; lexical overlap misses paraphrases; long-form QA contains multiple facts.

Research question: can embedding-space similarity serve as a correctness proxy?

Our hypothesis: raw cosine is weak, but refined statement-level similarity may be predictive.

2. Theory

Let p_i be the prediction statement and r_i the reference statement. A sentence encoder $E(\cdot)$ produces the cosine score

$$s_i = \cos(E(p_i), E(r_i)) = \frac{E(p_i) \cdot E(r_i)}{\|E(p_i)\| \|E(r_i)\|}.$$

We predict correctness with a threshold classifier:

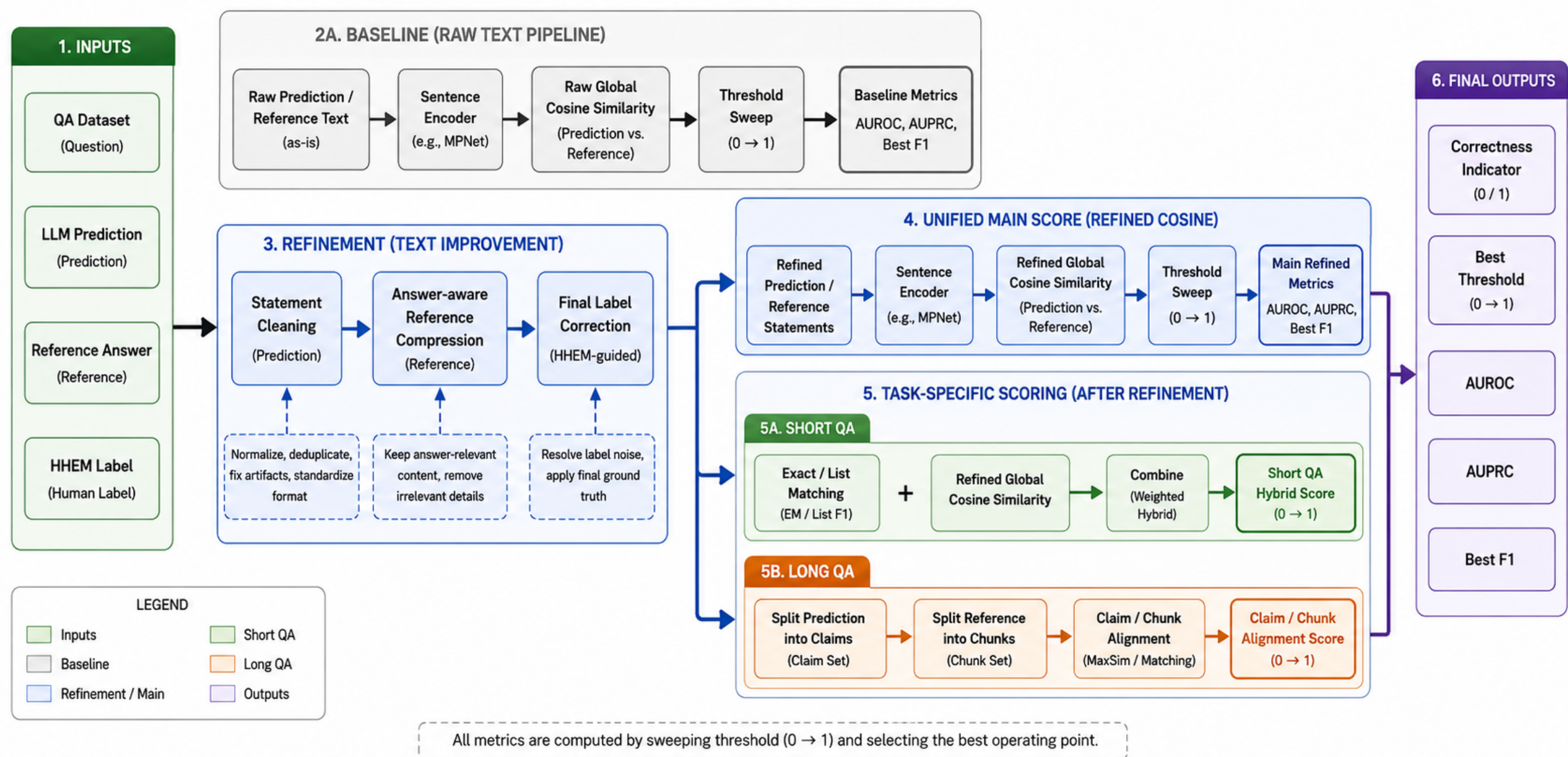
$$\hat{y}_i(\tau) = \mathbb{I}[s_i \geq \tau].$$

We sweep $\tau \in [0, 1]$ with step 0.01 and report AUROC, AUPRC, and Best F1. On the raw baseline, the cosine–label correlation is close to zero on the short datasets, which explains why refinement is necessary.

Intuition: Sentence encoders cluster semantically equivalent statements in latent space. Thresholding cosine induces a ranking-based correctness proxy. Long-form QA breaks this assumption because one global vector must summarize multiple claims.

3. Solution: Method Pipeline

Task-aware Semantic Similarity Evaluation Pipeline



4. Experiments

4.1 Data Statistics

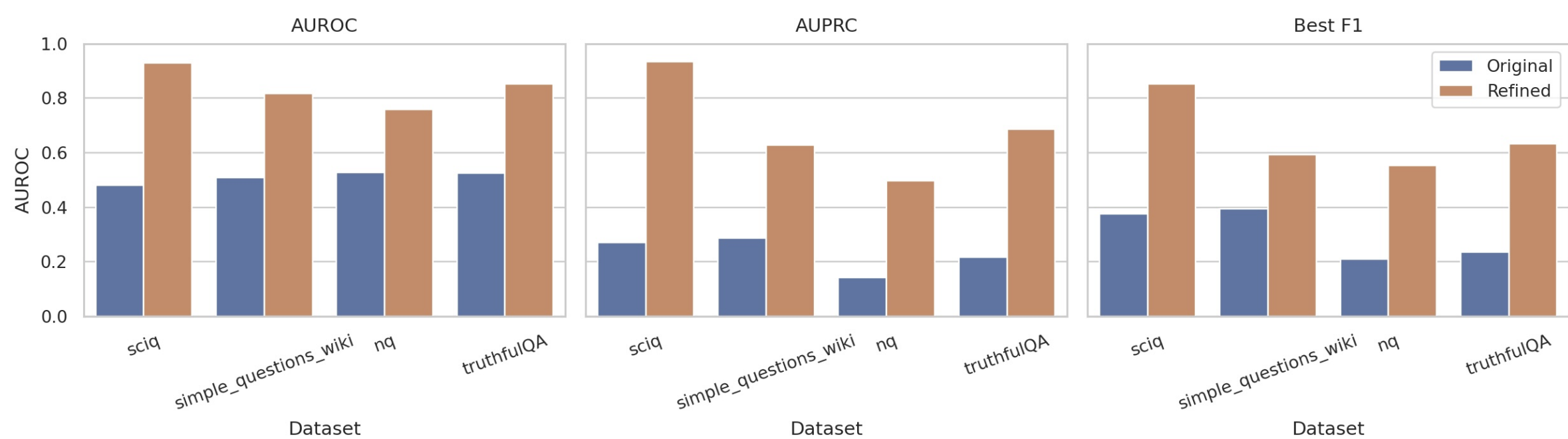
Main-subset experiments use seeded shuffle sampling.

Dataset	Type	n	Positive rate
SciQ	short	1000	42.5%
Simple Questions Wiki	short	1000	8.7%
Natural Questions	long	1000	26.2%
TruthfulQA	long	817	11.6%

4.2 Processing Procedures

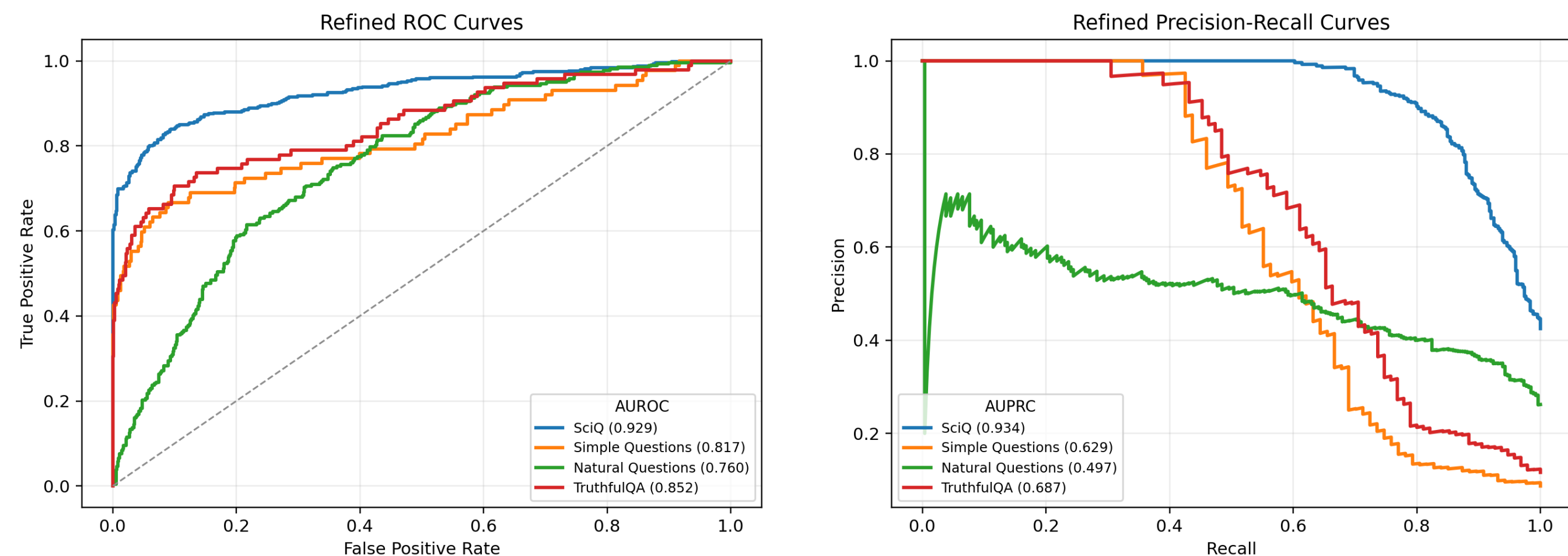
- Deterministic decoding: temperature 0.0, top-p 1.0.
- Max new tokens: 24 for short answers, 128 for long answers.
- Embedding batch size: 64; max length: 384.
- Evaluation: threshold sweep plus ranking metrics.

4.3 Main Experimental Results

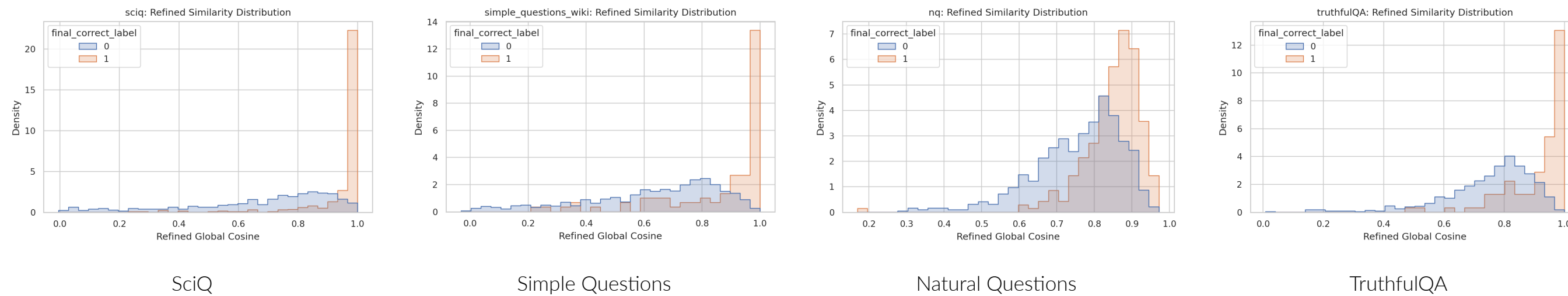


Base: raw cosine with original HHEM labels. **Refined:** cleaned statement cosine with **final_correct_label**. Higher AUROC/AUPRC means correct answers rank above incorrect ones.

Dataset	base AUROC	refined AUROC	base AUPRC	refined AUPRC	base F1	refined F1
SciQ	0.482	0.929	0.270	0.934	0.377	0.852
Simple Questions	0.510	0.817	0.287	0.629	0.394	0.592
Natural Questions	0.528	0.760	0.143	0.497	0.209	0.552
TruthfulQA	0.525	0.852	0.216	0.687	0.236	0.632



AUROC: four datasets in one ROC plot
Refined cosine consistently improves ranking quality and thresholded Best F1.



4.4 Short vs. Long Answer Behavior

Answer type	Positive rate	AUROC	AUPRC	Best F1
Short	25.6%	0.918	0.886	0.810
Long	19.6%	0.776	0.509	0.498

Short QA has clearer similarity separation. Long QA is harder because answers can be partially correct across multiple claims.

5. Ablation Studies

Ablated components. We compare raw cosine, label-only cleanup, unified refined cosine, and task-specific scores. Short hybrid combines exact/list matching with refined cosine. Claim/chunk alignment decomposes long answers into local factual units.

	AUROC				AUPRC				Best F1			
Raw cosine	0.482	0.510	0.528	0.525	0.270	0.287	0.143	0.216	0.377	0.394	0.209	0.236
Label only	0.764	0.862	0.573	0.627	0.701	0.593	0.307	0.311	0.666	0.233	0.431	0.319
Semantic refine	0.927	0.817	0.760	0.845	0.930	0.629	0.497	0.659	0.849	0.592	0.552	0.609
All refinements	0.929	0.817	0.760	0.852*	0.934	0.629	0.497	0.687*	0.852	0.592	0.552	0.632
Claims/chunk	0.929	0.825*	0.772*	0.848	0.934	0.629	0.506*	0.681	0.852	0.593	0.560*	0.634*
Short hybrid	0.915	0.795	N/A	N/A	0.929	0.442	N/A	N/A	0.828	0.439	N/A	N/A
Hybrid w/o exact	0.915	0.795	N/A	N/A	0.920	0.442	N/A	N/A	0.828	0.434	N/A	N/A
Hybrid w/o token-F1	0.933*	0.818	N/A	N/A	0.939*	0.635*	N/A	N/A	0.857*	0.604*	N/A	N/A
	SciQ	SimpleQ	NQ	TruthfulQA	SciQ	SimpleQ	NQ	TruthfulQA	SciQ	SimpleQ	NQ	TruthfulQA

* best per dataset and metric; N/A = short-only variant not applied to long QA

Dataset	Task-specific score	Refined AUROC/AUPRC/F1	Task AUROC/AUPRC/F1
SciQ	Short hybrid	0.929/0.934/0.852	0.933/0.939/0.857
SimpleQ	Short hybrid	0.817/0.629/0.592	0.818/0.635/0.604
NQ	Claim/chunk	0.760/0.497/0.552	0.772/0.506/0.560
TruthfulQA	Claim/chunk	0.852/0.687/0.632	0.848/0.681/0.634

Takeaway. Unified refinement is the main pipeline. Task-specific scoring is the stronger final policy: short hybrid improves short-answer F1, while claim/chunk alignment improves long-answer F1 and NQ ranking.

6. Case Studies: Error Diagnosis

Case	What happens	Method response
Paraphrase	Same fact, different surface form.	Refined cosine should pass the threshold.
Short label noise	Exact short answer marked wrong by HHEM.	Short hybrid and label correction recover it.
Partial long answer	One claim is right, another is missing.	Claim/chunk alignment avoids one-vector overconfidence.
Prompt leakage	Copied Q&A fragments dilute the answer.	Statement cleaning reduces false negatives.

7. Conclusions

- Similarity is a useful threshold-based correctness signal after refinement, not as raw cosine.
- Unified refinement lifts AUROC from near-random 0.48–0.53 to 0.76–0.93.
- Best implemented policy: short hybrid for short QA; claim/chunk alignment for long QA.
- Remaining issue: calibrating thresholds for imbalanced and partially correct long answers.